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***In vivo* spectroscopy and internal optics of leaves as basis for remote sensing of vegetation**

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Abstract. *In vivo* reflection spectra of intact bean leaves (*Phaseolus vulgaris*) were measured between 400 and 800 nm under remote sensing conditions (illumination with white light, detection of a narrow angle of the reflected light) using the VIRAF spectrometer. The leaves with colours from yellow to green were chosen at different times during light-induced greening. The colours of the leaves were characterized by the chromaticity coordinates according to CIE 1931 calculated from the reflection spectra. The influence of the absorption of chlorophyll—the main pigment of green leaves—on the reflection spectrum of leaves is outlined. The shape of the *in vivo* reflection spectra is interpreted taking into account (a) the formation of pigment-protein complexes, (b) the sieve effect and the detour effect, as well as (c) the reflection, refraction and scattering of light inside the leaf tissue. Reflection signals at several distinct wavelengths and their ratios as well as the inflection point of the reflection rise from the far red towards the near-infrared were checked for linear correlation with the chlorophyll content per leaf area. The normalized difference vegetation index (NDVI) exhibited a relatively bad correlation with the chlorophyll content. The best correlation was found for the logarithm of the ratio of the reflection signals at 800 and 550 nm. We suggest that this parameter is used, in combination with the detection of the inflection point, for remotely-detecting vegetation and for the estimation of its chlorophyll content.

1. Introduction

Remote sensing of vegetation in the visible and near-infrared is used to detect the distribution, health and productivity of plants. For this purpose reflectance signals are usually measured in the green, red, and far red region of the spectrum. By means of calculating signal ratios, e.g., the normalized difference vegetation index NDVI, vegetation can be distinguished from non-vegetation (Rouse *et al.* 1974) and corn yield could be predicted (Tucker *et al.* 1980). The shift of the steep rise of the reflection spectrum from the far red towards the near-infrared (red edge) is taken as an indicator of stress or damage to plants (Gates *et al.* 1965, Chang and Collins 1983). From the reflection signals the absorbed photosynthetic active radiation aPAR has been deduced (Asrar *et al.* 1984) and aPAR accumulated over the vegetation period has been taken as a measure for the productivity of agricultural plants (Sellers 1985, Sellers 1987). Biomass production has also been correlated with the reflection signal at only one distinct wavelength range (Tucker 1979).

It is well established that the absorption of leaf pigments, with the green chlorophyll as main component, determines the shape of the reflection spectra. However, the particular spectroscopic and optic characteristics of the leaf tissue are usually not considered when interpreting remotely-sensed data of vegetation. Here

we present *in vivo* reflection spectra of leaves with colours varying from yellow to fully-green measured in the laboratory using the VIRAF spectrometer. This apparatus was constructed by our working group for measuring not only reflection spectra but also absorption and fluorescence emission spectra of one sample in order to receive better and comparable spectroscopic information of intact leaves (Buschmann *et al.* 1988, 1991). The *in vivo* spectra are interpreted taking into account the internal optics of the leaf which comprise reflection, refraction and scattering of light inside the leaf tissue (Butler 1964, Woolley 1971, Vogelmann 1989). The correlation of different reflection signals and/or signal ratios with the chlorophyll content per leaf area has been calculated in order to optimize the use of these signatures for remote sensing of vegetation and for estimation of its chlorophyll content.

2. Material and methods

Seeds of bean (*Phaseolus vulgaris*, var. Fori) were soaked in water for 2 hours and then placed on peat. After 8 days of growth in complete darkness the plants were transferred into white light ($150 \mu\text{Em}^{-2}\text{s}^{-1}$) for up to 20 days. Leaves with different chlorophyll content were taken at different times during greening in the light. Three types of brown soil with varying consistence were taken from different sites of the campus of the University of Karlsruhe.

The reflection spectra were taken between 400 and 800 nm in steps of 2 nm using the VIRAF spectrometer (VIRAF=visible infrared reflection absorption fluorescence) developed by our working group (Buschmann *et al.* 1988). In contrast to conventional reflection spectrometer in the VIRAF spectrometer the sample is illuminated, like in remote sensing, with white light instead of monochromatic light. The light of a halogen lamp reaches the sample rectangularly on the upper leaf side after reflection at a 50 per cent transparent mirror. As in remote sensing, only a narrow angle of the reflected light is detected. The reflected light passes the 50 per cent transparent mirror in a straight direction in order to reach a $0^\circ/0^\circ$ geometry. The light beam is analysed by means of a monochromator (H25 Jobin Yvon, Instruments SA (ISA), D-8011 Grasbrunn, Germany) with a grating of 1200 lines per mm. The spectral dispersion was adjusted to 9 nm. A photomultiplier (R 406 Hamamatsu, Toyooka, Japan) was used to measure the monochromatic light at the exit slit of the monochromator. The scanning of the monochromator and the voltage adjustment of the photomultiplier were controlled via an interface (Spektralink, Jobin Yvon, ISA) connected to a microcomputer which also receives the measuring signals. As reference sample for reflection spectra a ceramic white standard (SRS-99 Spektralon, LOT, D-6100 Darmstadt, Germany) with more than 99 per cent reflectance between 400 and 1700 nm was used. The calculation of the spectra and the control of the monochromator and the photomultiplier was carried out by means of commercially available software (Enhanced Prism Software, Jobin Yvon, ISA). The correlation of the reflection data with the chlorophyll content was calculated using LOTUS 123 software (LOTUS Corporation, Berkshire SL4 1EX, England). The chromaticity coordinates according CIE 1931 (MacAdam 1981) were calculated from the *in vivo* reflection spectra of leaves for a light source type D₆₅ using a self-written BASIC programme.

The *in vivo* reflection spectra of some leaves were measured before and after infiltration with water under vacuum. The reflection spectra of other leaves were taken before extracting the leaf pigments with acetone. For measuring the reflection spectra of leaf extracts the extract was loaded on a thin layer chromatography plate

(silica gel 60, Nr. 5749, Merck, D-6100 Darmstadt, Germany) and the spectra were taken after evaporation of the solvent. The reflection of soil was detected after fixing the samples on a support by means of an adhesive.

The chlorophyll content of the leaves was determined immediately after measuring the *in vivo* reflection spectra with the VIRAF spectrometer. The leaf pigments were extracted with acetone and the content of chlorophyll *a* and *b* as well as of carotenoids, was calculated using the coefficients of Lichtenthaler (1987).

3. Results

3.1. Chromaticity coordinates of leaves

A leaf grown in the dark has no chlorophyll and therefore shows the yellow colour of carotenoids. During illumination chlorophyll is gradually accumulated and the leaf turns green (review: Sesták 1985). This light-induced greening process can be followed by the chromaticity coordinates according CIE 1931 which are calculated from *in vivo* reflection spectra. Both *x* and *y* coordinates decrease with increasing chlorophyll content per leaf area (figure 1 and table 1). Concomitantly, the lightness of the leaf decreases and the dominant wavelength is shifted from the yellow around 570 nm towards the blue-green around 500 nm (figure 1 and table 1).

3.2. In vivo reflection spectra of leaves

In vivo reflection spectra of leaves are characterized by a low signal in the blue spectral region between 400 and 500 nm and a high signal in the near-infrared between 750 and 800 nm (figure 2, upper part). With increasing chlorophyll content a negative band around 680 nm appears, i.e., the reflection signal in this region decreases. The minimum of the 680 nm reflection is already reached at 10 μg chlorophyll per sq. cm. With increasing chlorophyll content the reflection at 680 nm remains constant. The most significant changes of the *in vivo* reflection spectrum

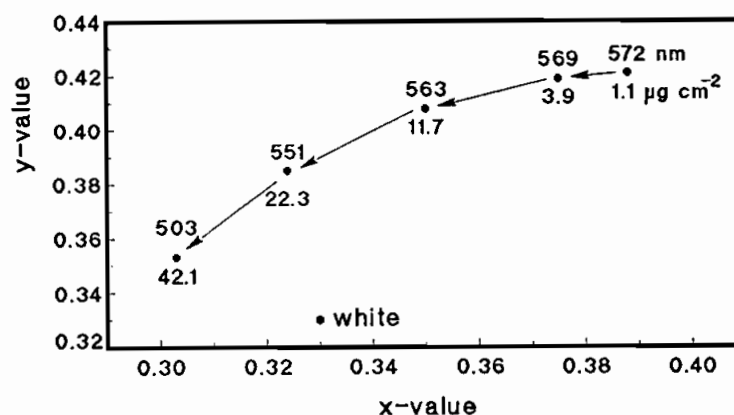


Figure 1. Chromaticity coordinates according to CIE 1931 of bean leaves with different chlorophyll content per leaf area calculated from the *in vivo* reflection spectra of leaves. Leaves were determined at different times during greening in the light. The time sequence of illumination is indicated by the arrows. The coordinates $x=0.33$ and $y=0.33$ represent the point for white colour (i.e., no colour). In addition, the chlorophyll content per leaf area in $\mu\text{g cm}^{-2}$ (upper value) and the dominant wavelength in nm (lower value) are given. Each point represents the mean of five samples measured at the same time during the greening process.

Table 1. Chromaticity coordinates according to CIE 1931 of bean leaves with different chlorophyll content per leaf area calculated from the *in vivo* reflection spectra.

Chlorophyll content [$\mu\text{g cm}^{-2}$]	Chromaticity coordinates according to CIE 1931		Lightness [%]	Dominant wavelength [nm]
	x-value	y-value		
1.1 ± 0.2	0.388	0.421	28.1	572
2.1 ± 0.3	0.371	0.408	25.1	570
3.9 ± 0.9	0.375	0.419	26.8	569
11.7 ± 1.3	0.350	0.408	17.7	563
12.2 ± 1.1	0.343	0.400	17.1	562
22.3 ± 2.0	0.324	0.385	12.9	551
27.8 ± 3.0	0.318	0.379	11.8	540
30.3 ± 4.7	0.322	0.393	14.5	547
34.5 ± 3.0	0.312	0.369	11.7	522
42.1 ± 5.0	0.303	0.353	11.2	503

The plants were grown in the dark for 8 days and the leaves were taken at different times during growth under white light (i.e., gradual light-induced greening). Mean values of five samples. Compare with figure 1.

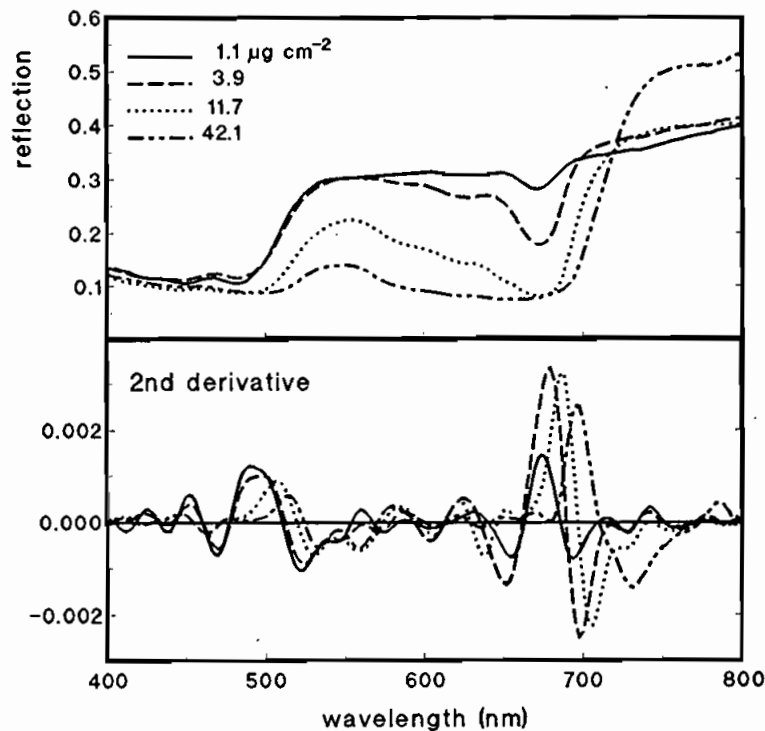


Figure 2. (upper part). *In vivo* reflection spectra of leaves with four different values for the chlorophyll content per leaf area. The illumination time for the greening process was 0 h, 1.5 h, 5.5 h and 20 days, respectively. (lower part). Second derivative of the reflection spectra given in the upper part. The intersection with the abscissa around 700 nm marks the inflection point of the rise of the original reflection spectrum from 680 nm towards longer wavelengths (red edge).

occur within the first 6 hours of greening in the light. This is shown in the three spectra of figure 2 (upper part) with the lowest chlorophyll content taken after 0 h, 1.5 h and 5.5 h illumination. The development to the spectrum with the highest chlorophyll content (figure 2, upper part) took about 20 days of illumination. During greening of the leaf the negative band at 680 nm becomes broader and the rise of the signal from 680 to 750 nm is shifted towards longer wavelengths. This shift can be detected by the shift of the inflection point around 700 nm determined from the intersection of the second derivative with the abscissa (figure 2, lower part). With increasing chlorophyll content the signal of the entire spectrum between 500 and 680 nm decreases. Finally, the dark-green leaf shows only a small peak in the green part of the spectrum around 550 nm (figure 2, upper part). This dark-green leaf is also characterized by a high reflection in the near-infrared. In general, the reflection at 800 nm rises gradually during greening (see equation in table 2).

3.3. Linear correlation of reflection signals with the chlorophyll content

When taking only the height of the reflection signal at one particular wavelength the best correlation with the chlorophyll content per leaf area showed up for the reflection signal at 550 nm (highest r^2 -value, table 2). An even better linear correlation was achieved when calculating the difference between the signals at 800 and 550 nm (figure 3 and table 2). The best fitting linear correlation, however, was

Table 2. Reflection signals of bean leaves with different chlorophyll content and their linear correlation with the chlorophyll content per leaf area. For comparison the range of reflection signals of different types of brown soil is given.

Reflection parameter	Leaf		Linear correlation			Soil (range)
	Minimum	Maximum	<i>a</i>	<i>b</i>	r^2	
R(476)	0.0694	0.1300	-0.00070	0.11016	0.474	0.061-0.183
R(550)	0.1330	0.3410	-0.00387	0.28524	0.756	0.055-0.220
R(632)	0.0699	0.3419	-0.00523	0.25362	0.741	0.059-0.261
R(680)	0.0628	0.3255	-0.00402	0.20510	0.564	0.073-0.284
R(800)	0.2850	0.5640	0.00336	0.38068	0.525	0.174-0.384
R(800)-R(476)	0.1680	0.4707	0.00406	0.27052	0.670	0.100-0.206
R(800)-R(550)	0.0510	0.4190	0.00723	0.09543	0.906	0.112-0.181
R(800)-R(632)	0.0691	0.4775	0.00859	0.12706	0.871	0.105-0.161
R(800)-R(680)	0.0855	0.4785	0.00738	0.17558	0.779	0.088-0.143
log R(800)/R(476)	0.3867	0.8044	0.00632	0.54239	0.792	0.739-1.121
log R(800)/R(550)	0.0691	0.6108	0.01116	0.12407	0.942	0.556-1.228
log R(800)/R(632)	0.0799	0.8662	0.01769	0.18051	0.908	0.386-1.154
log R(800)/R(680)	0.1013	0.8444	0.01515	0.30692	0.742	0.302-0.935
R(476)/R(800)	0.1569	0.4105	-0.00332	0.28688	0.728	0.326-0.478
R(550)/R(800)	0.2451	0.8529	-0.01228	0.72804	0.884	0.293-0.574
R(632)/R(800)	0.1361	0.8319	-0.01458	0.64491	0.798	0.315-0.680
R(680)/R(800)	0.1431	0.7920	-0.01156	0.52586	0.641	0.393-0.739
NDVI	0.1161	0.7497	0.01236	0.32961	0.692	0.150-0.436
IP	686	712	0.55591	688.534	0.929	—

$R(n)$ =reflection signal at the wavelength n , NDVI=normalized difference vegetation index $= (R(800) - R(680)) / (R(800) + R(680))$, IP=inflection point of the rise of the reflection spectrum from 680 nm towards the near-infrared, a and b =parameters of the equation for a linear correlation: $y = ax + b$, r^2 =correlation coefficient.

found with the logarithm of the ratio between the signals at 800 and 550 nm (figure 4 and table 2). The inflection point of the red edge also gives an excellent correlation with the chlorophyll content (figure 5 and table 2). In contrast to this, the normalized difference vegetation index NDVI showed only a relatively poor linear correlation ($r^2=0.692$, figure 6 and table 2). For the NDVI a linear correlation could only be found at a low chlorophyll content per leaf area. However, with chlorophyll concentrations higher than $10 \mu\text{g cm}^{-2}$ the NDVI shows only slight changes.

3.4. Reflection spectrum of a leaf extract and of soil

The reflection spectrum of a leaf extract shows a higher signal in the spectral region between 500 and 650 nm (figure 7). Compared to the *in vivo* spectrum of the yellowish-green leaf (figure 2, upper part) the peak at 550 nm is even more pronounced. The spectrum of the extract shows a small minimum around 440 nm which cannot be detected in the spectrum of the intact leaves. Furthermore the negative band around 680 nm is slightly shifted towards shorter wavelengths. The reflection intensity of brown soil samples is low (figure 7). The spectrum is nearly flat showing a slow rise towards longer wavelengths. The reflection signals and signal ratios of different types of brown soil with varying consistence showed values which overlap with, or lay in, the same minimum-maximum range as those of the bean leaves (table 2).

3.5. In vivo reflection spectrum of a water-infiltrated leaf

Water infiltration leads to a decrease of the *in vivo* reflection signal, in particular in those parts of the spectrum of a leaf which are normally characterized by a high reflection: i.e., the green to red spectral region from 500 to 650 nm and the near-infrared from 700 to 800 nm (figure 8).

4. Discussion

The *in vivo* reflection spectra of green leaves are mainly influenced by the absorption characteristics of chlorophyll. Absorption maxima in the far red (around

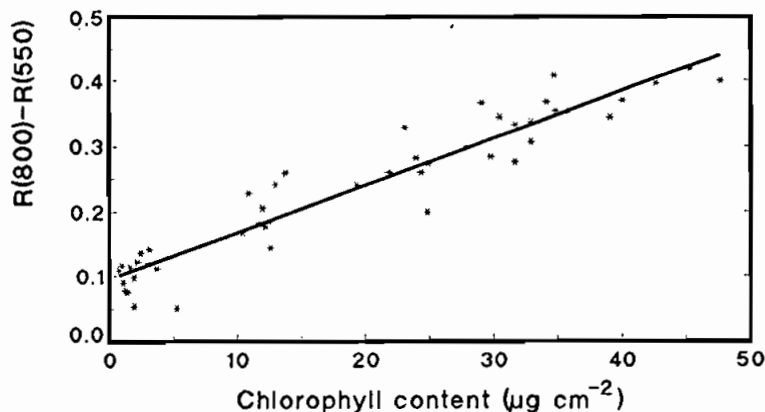


Figure 3. Correlation between the difference of the reflection signals at 800 and 550 nm with the chlorophyll content per leaf area. The result of the calculation of a linear correlation is indicated by the full line. The equation for this line is given in table 2. The data were taken from the *in vivo* reflection spectra of 47 leaves with different chlorophyll content.

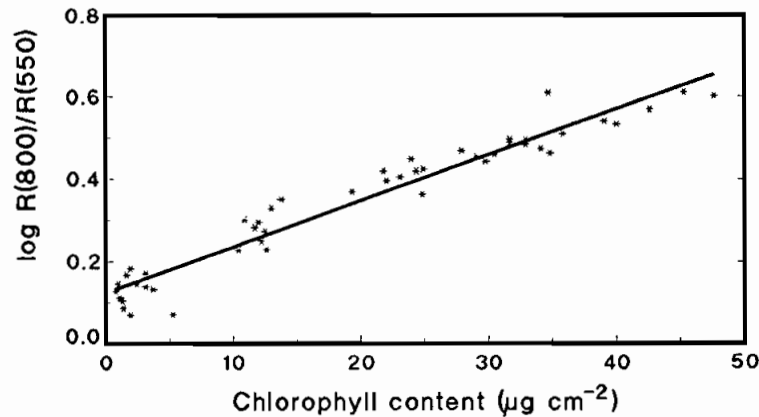


Figure 4. Correlation between the logarithm of the ratio between the reflection signals at 800 and 550 nm with the chlorophyll content per leaf area. Otherwise as for figure 3.

680 nm) and in the blue (around 450 nm) are reflected in the reflection minima. Absorption minima in the green (around 550 nm) and in the near-infrared (above 750 nm) correspond to reflection maxima (figure 2, upper part, as well as figures 7 and 8). In the spectrum of the yellow leaf (figure 2, upper part) the contribution of carotenoid absorption between 400 and 500 nm which is also present in green leaves can be seen (review: Sesták 1985).

With increasing chlorophyll concentration the reflection signal decreases in the far red absorption maximum of chlorophyll, i.e., at 680 nm. However, this decrease which is expected from the rise of absorption according to the Lambert Beer's law can only be observed at the onset of greening. At chlorophyll concentrations higher than $10 \mu\text{g cm}^{-2}$ the reflection minimum remains at a constant low level. The constant reflection minimum is caused by the internal optics of the leaf tissue, namely the sieve effect and the detour effect (Rabinowitch 1951, Vogelmann 1989). The sieve effect is observed when the absorbing pigments are not homogeneously distributed in the sample. Chlorophylls and most carotenoids are exclusively located in the internal

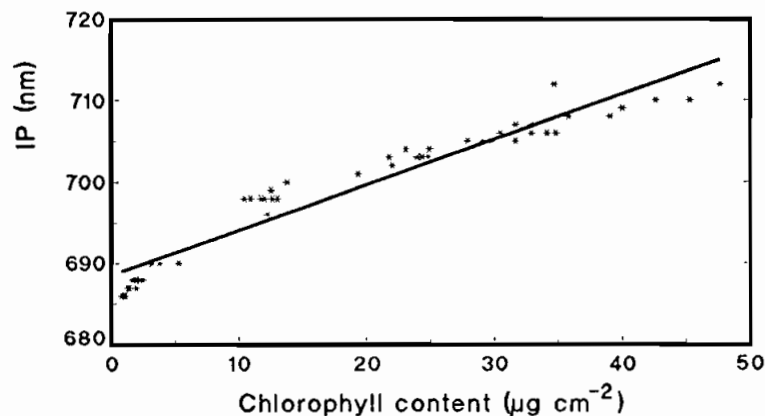


Figure 5. Correlation between the inflection point IP of the red edge and the chlorophyll content per leaf area. Otherwise as for figure 3.

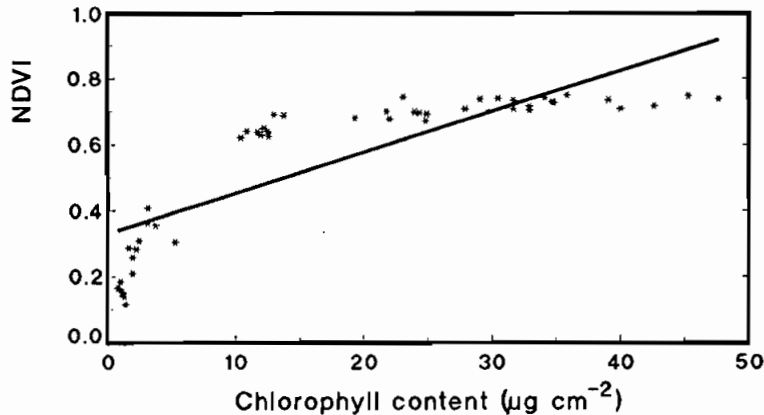


Figure 6. Correlation between the normalized difference vegetation index NDVI and the chlorophyll content per leaf area. Otherwise as for figure 3.

membranes of the chloroplasts, i.e., small particles (organelles) typical for plant cells. The sieve effect leads to a decrease (or a decreased rise) of the absorption maxima and thus to the broadening of the absorption bands which can be seen during the greening process as the broadening of the negative reflection band at 680 nm (figure 2, upper part). The detour effect is caused by the lengthening of the light path through the leaf tissue induced by internal reflection, refraction and scattering. Radiation in the visible and near-infrared easily penetrates the colourless, outer cell layer of a leaf (the epidermis, figure 9). In the green tissue between the two outer cell layers (the mesophyll) light is not only absorbed or transmitted but also deviated from its direct way to the lower leaf side by scattering, reflection and refraction. The modification of the straight light path through the leaf has been theoretically predicted and calculated (Willstätter and Stoll 1918, Gates *et al.* 1965, Seyfried and Fukshansky 1983). It has been measured in the past by inserting micro fiber optic probes into leaves (Vogelmann and Björn 1984, Vogelmann 1989). Intercellular air spaces and organelles (e.g.,

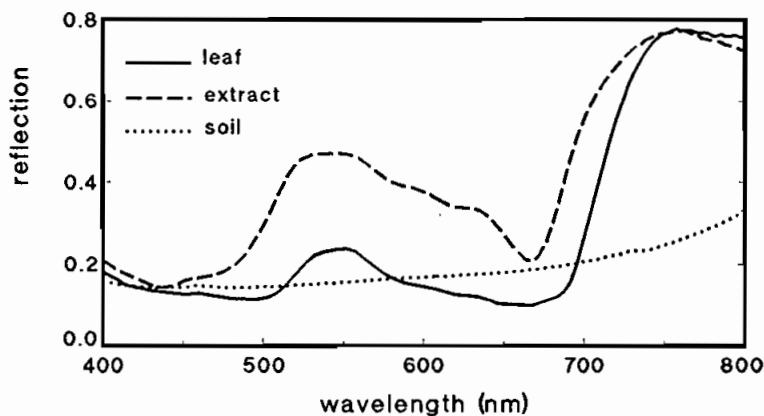


Figure 7. *In vivo* reflection spectrum of a green bean leaf and reflection spectrum of its pigments extracted with acetone. The spectrum of the extract is normalized at 750 nm to the same height as the spectrum of the leaf. For comparison the reflection spectrum of a brown soil is given.

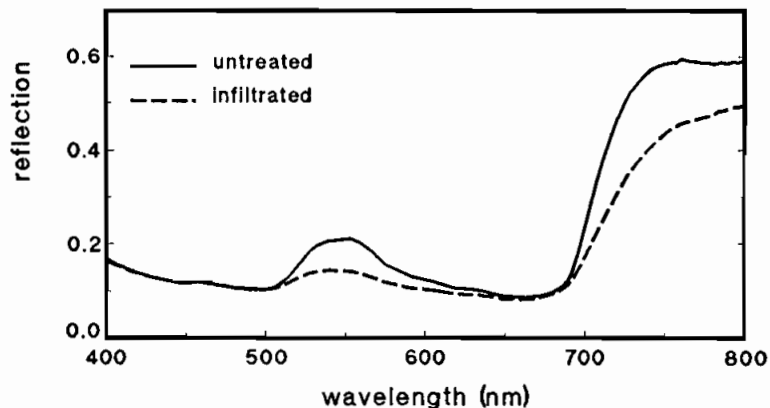


Figure 8. *In vivo* reflection spectrum of a green bean leaf before and after vacuum infiltration with water.

chloroplasts) are the main cause for light scattering inside the leaf. Scattering in a leaf is mainly Mie-scattering (Vogelmann 1989), which occurs at particles with diameters somewhat larger than the wavelength of the incident light (Gobrecht 1974). In contrast to the Rayleigh-scattering at particles with diameters smaller than the wavelength of the incident light, Mie-scattering is wavelength-independent and distributes light preferentially in the forward direction (Gobrecht 1974). Reflection and refraction occurs at borders between phases with different refractive indices, e.g., at the border between cells (refractive index of about 1.4, Gausman *et al.* 1974) and the intercellular air spaces (refractive index of air: 1.0). The lengthening of the light path through the leaf by internal reflection, refraction and scattering leads to an increased probability for light absorption. This detour effect is characterized by an increased absorption in the range of the absorption minima, e.g., between 500 and 650 nm and above 680 nm. The increased absorption is reflected by the decreased reflection in these spectral ranges (figure 2, upper part).

Another effect of the internal optics of the leaf can be observed at 800 nm. The reflection signal gradually rises during the greening process. In this part of the spectrum chlorophyll does not absorb anymore. The reflection in this spectral range becomes higher because straight penetration of light inside the leaf tissue is more disturbed by scattering and/or reflection and refraction. A decrease of reflectance in this spectral range has been detected for damaged leaves and was attributed to anatomical and morphological disturbance of the leaf tissue structure (Hoffer and Johannsen 1969). Intercellular air spaces and cell walls are known to be changed during the development and growth of the leaf. The number and size of the intercellular air spaces increases with the ageing of the leaves. Filling the intercellular air spaces of a fully developed green leaf by infiltrating with water shows the inverse effect: i.e., the decrease in the reflection at 800 nm (figure 8) which is paralleled by a higher transmittance of light in this spectral range.

Unlike the situation in a solution, the chlorophyll in a leaf is non-covalently bound to proteins forming different pigment-protein-complexes (French *et al.* 1972). By extracting the leaf pigments with acetone these complexes are destroyed, the chlorophyll is dissolved and the proteins are coagulated. Due to the formation of pigment-protein-complexes the absorption maximum of chlorophyll is shifted

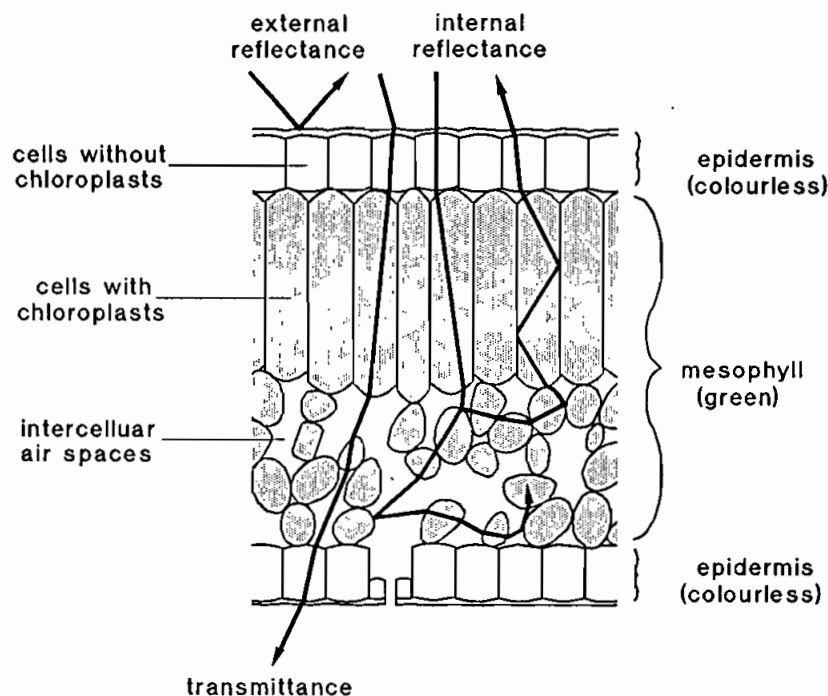


Figure 9. Scheme of a cross-section of a leaf with the light incident at the upper leaf side. The lines indicate external and internal reflectance as well as transmittance showing possible ways of deviation of the light from its straight path through the leaf tissue.

towards longer wavelengths. Since several pigment-protein-complexes occur, each with a slightly different absorption maximum, the absorption band in the far red is broader for the intact leaf than for the extract which has only one major absorption band for chlorophyll. The absorption band of a fully transparent extract is still less broad than the negative band of the reflection spectrum of the highly light scattering silica gel plate used in this investigation (Buschmann and Nagel 1991).

The normalized difference vegetation index (NDVI) (Rouse *et al.* 1974) is calculated using the reflection values at lowest chlorophyll absorption (in the near-infrared) and highest chlorophyll absorption (in the far red). However, as is shown in figure 6, a linear correlation with the chlorophyll content holds only at values below $10 \mu\text{g cm}^{-2}$. Therefore the NDVI could be useful to distinguish vegetation from non-vegetation but should not be applied for determining the chlorophyll content of leaves. At low values of NDVI the signal may get close to the value calculated for soil (table 2). Furthermore the NDVI may strongly be affected by reflectance which is constant over the entire spectrum, e.g., background soil and/or waxes and hairs of the leaf surface.

The determination of absorbed photosynthetic active radiation is based on correlation with the NDVI (Asrar *et al.* 1984, Sellers 1985, 1987). Therefore the same restrictions as for the NDVI itself are valid. A high chlorophyll concentration guarantees a high light absorption, however, it does not necessarily mean a high photosynthetic activity and a high biomass production. Different plants and even different leaves of one plant (e.g., a tree) are known to be characterized by different efficiency of photosynthesis and plant growth (see e.g., Mitsui and Black 1982).

The inflection point of the red edge (Gates *et al.* 1965, Chang and Collins 1983) is well correlated with the chlorophyll content of the leaf. This can be explained by the broadening of the band of the chlorophyll absorption maximum at 680 nm during the greening process, which is due to the increased absorption but also to the sieve and detour effect (see above). Since only the shape of the reflection spectrum is considered when calculating this inflection point, the contribution of a reflectance which is constant throughout the spectrum is minimized (in contrast to the NDVI, see above). In addition, there is no inflection point in this spectral region for the reflection spectrum of the soil (figure 7). Unfortunately, the determination of the inflection point needs a high spectral resolution which makes it impractical for conventional, low-spectral-resolution remote sensing.

The usual detection of distribution, health and productivity of plants by remote sensing in the visible and near-infrared is based on the determination of the chlorophyll concentration at a particular site. One should, however, be aware of the fact that by means of this technique damage, stress effects and loss of productivity can only be sensed when a decrease of chlorophyll is observed. In case other parameters are effected remote sensing must be carried out using different techniques, e.g., laser-induced fluorescence and/or detection of microwaves or signals in the thermal infrared. For remote detection of vegetation we suggest to use the logarithm of the ratio between the reflection signals at 800 and 550 nm. This parameter obviously has a high linear correlation with the chlorophyll content per leaf area which is needed for a clear determination of vegetation and even a quantitative monitoring of chlorophyll content. The relatively large spectral dispersion (9 nm) used in our measurements make these results comparable to the broad-band AVHRR data.

With exception of the inflection point, each reflection signal or signal ratio of the brown soil types lay in the range of the minimum and maximum of the reflection signal of the leaf (table 2). Therefore the existence of the inflection point around 700 nm should be checked in addition to the determination of $\log R_{800}/R_{680}$ in order to exclude the detection of soil. Like for the other more established parameters this logarithmic ratio is expected to vary from plant to plant due to differences in the external and internal optics of the particular leaf. The relatively good linear correlation of the signal at 550 nm holds also for the value at 632 nm. This shows that measuring with broad bands or channels should also lead to good results.

Acknowledgment

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